Bounded no-in-house sequential fractional procurement: A finite-dimensional reduction and the two-seller value

Draft note

June 23, 2026

Abstract

Qin, Vardi, and Wierman left open the problem of analytically resolving the exact competitive ratio for minimization fractional prophet inequalities without in-house production, for arbitrary exponent $p > 1$, number of sellers n, and bounded cost support $[\ell, u]$. This note gives partial progress on that problem. Under the standard prophet-inequality convention that the online decision maker knows the product distribution of costs before choosing the optimal policy for that instance, the infinite-dimensional supremum over all product distributions supported on $[\ell, u]$ is reduced exactly to a compact optimization over one point mass in the first coordinate and two-point laws in all later coordinates. Thus the general problem becomes a finite-dimensional, but still nonconvex, maximization problem. In the special case $n = 2$, the reduced problem is solved in closed form for every $p > 1$, extending the known $n = p = 2$ formula. The results here should therefore be read as a complete solution of the two-seller case and as substantial partial progress, not as a closed-form solution of the full problem for all n.

1 Introduction and scope

Qin, Vardi, and Wierman [1] study a sequential procurement problem in which a firm must buy one unit of a divisible commodity from sellers arriving in a fixed order. Seller i has realized cost coefficient $c_i > 0$, and if the firm buys amount $x_i \geq 0$ from seller i, then the incurred cost is $c_i x_i^p$. The decisions are irrevocable and must satisfy $\sum_i x_i = 1$. A prophet sees all realized costs before allocating demand, while the online decision maker observes costs sequentially. In the no-in-house version, every coordinate is an arriving seller rather than a deterministic in-house production option.

The article [1] solves several regimes and gives the exact no-in-house formula in the special case $n = p = 2$. It also explains that analytically resolving the exact ratio for the bounded no-in-house case for general p, n, ℓ, u is more difficult. The present note addresses precisely this bounded no-in-house case.

The main point is a reduction in the distributional degrees of freedom. Let $\Gamma_{p,n}(\ell, u)$ denote the worst-case ratio when, for each product law D, the online policy is chosen optimally using knowledge of D. The first result proves that

$$\Gamma_{p,n}(\ell, u)$$

can be computed by optimizing over finitely many variables: the first distribution may be taken to be a point mass, and every later distribution may be taken to have support of size at most two. This is exact, but it is not the same as a closed-form evaluation for general n. For $n \geq 3$, the resulting finite-dimensional problem remains nonconvex and involves 2^{n-1} prophet terms.

The second result solves the reduced problem when $n = 2$. Writing

$$\rho := \frac{u}{\ell}, \qquad \alpha := \frac{1}{p-1},$$

we obtain

$$\Gamma_{p,2}(\ell,u)=\frac{1+\sqrt{\rho}}{2\sqrt{\rho}}\left(\frac{1+\rho^{1/(2(p-1))}}{2}\right)^{p-1}.$$

For $p=2$ this becomes

$$\frac{1}{4}\left(\sqrt{\frac{u}{\ell}}+\sqrt{\frac{\ell}{u}}\right)+\frac{1}{2},$$

which is the two-seller quadratic formula of [1].

2 Model and preliminary identities

Fix $p>1$ and define

$$\alpha:=\frac{1}{p-1}.$$

For positive numbers $a_1,\dots,a_m$, define their p-parallel sum by

$$a_1\oplus\cdots\oplus a_m:=\left(\sum_{j=1}^{m}a_j^{-\alpha}\right)^{-1/\alpha}. \tag{1}$$

This operation is commutative, associative, increasing in each coordinate, and homogeneous of degree one.

Lemma 2.1 (Prophet value)**.** *For a realized cost vector $c=(c_1,\dots,c_n)\in(0,\infty)^n$,*

$$\mathrm{OPT}(c,p):=\min\left\{\sum_{i=1}^{n}c_ix_i^p : x_i\ge 0,\ \sum_{i=1}^{n}x_i=1\right\}=c_1\oplus\cdots\oplus c_n.$$

The unique minimizing allocation is

$$x_i=\frac{c_i^{-\alpha}}{\sum_{j=1}^{n}c_j^{-\alpha}},\qquad i=1,\dots,n.$$

Proof. The objective is strictly convex on the simplex. The Lagrange equations at an interior optimum are

$$pc_ix_i^{p-1}=\lambda,\qquad i=1,\dots,n.$$

Thus x_i is proportional to $c_i^{-1/(p-1)}=c_i^{-\alpha}$. Substitution gives the displayed value. Since every c_i is positive, the optimizer is interior and unique. □

Let $D=D_1\otimes\cdots\otimes D_n$ be a product law on $[\ell,u]^n$, where $0<\ell\le u<\infty$. The optimal online cost-to-go has a homogeneous form.

Lemma 2.2 (Online dynamic program)**.** *Define $B_i(D)$ backwards by*

$$B_n(D):=\mathbb{E}C_n,\qquad B_i(D):=\mathbb{E}_{C_i\sim D_i}\big[C_i\oplus B_{i+1}(D)\big],\quad i=n-1,\dots,1. \tag{2}$$

Then the minimum expected online cost for law D is $B_1(D)$. More generally, if the remaining demand before seller i is y, then the optimal expected cost-to-go is $B_i(D)y^p$. At a nonlast seller $i<n$, after observing $C_i=c$, an optimal purchase amount is

$$x_i=\frac{B_{i+1}(D)^{\alpha}}{c^{\alpha}+B_{i+1}(D)^{\alpha}}\,y=\frac{c^{-\alpha}}{c^{-\alpha}+B_{i+1}(D)^{-\alpha}}\,y. \tag{3}$$

At the last seller, all remaining demand is bought.

Proof. The claim is immediate at $i = n$. Suppose it holds at $i + 1$. If the remaining demand before seller i is y and the realized cost is c, then the decision maker minimizes

$$cx^p + B_{i+1}(D)(y - x)^p, \qquad 0 \le x \le y.$$

By the same one-step Lagrange calculation as in Lemma 2.1, the minimum is

$$\bigl(c \oplus B_{i+1}(D)\bigr)y^p,$$

and the minimizing x is (3). Taking expectation over C_i gives (2) and completes the backward induction. □

The distribution-aware bounded no-in-house value studied in this note is

$$\Gamma_{p,n}(\ell, u) := \sup_{D_1,\dots,D_n \in \mathcal{P}([\ell,u])} \frac{B_1(D)}{\mathbb{E}\bigl[C_1 \oplus \cdots \oplus C_n\bigr]}. \tag{4}$$

By homogeneity of $\oplus$,

$$\Gamma_{p,n}(\ell, u) = \Gamma_{p,n}(1, u/\ell). \tag{5}$$

3 An exact finite-dimensional reduction

Let

$$\mathcal{X}_n(\ell, u) := [\ell, u] \times \bigl([\ell, u]^2 \times [0, 1]\bigr)^{n-1}.$$

A point

$$x = \bigl(a_1, (a_2, b_2, q_2), \dots, (a_n, b_n, q_n)\bigr) \in \mathcal{X}_n(\ell, u)$$

represents the product law

$$C_1 = a_1 \quad \text{a.s.}, \qquad C_i = \begin{cases} a_i, & \text{with probability } q_i, \\ b_i, & \text{with probability } 1 - q_i, \end{cases} \qquad i = 2, \dots, n. \tag{6}$$

Degenerate cases are allowed.

For $x \in \mathcal{X}_n(\ell, u)$, define $\mathsf{B}_i(x)$ backwards by

$$\mathsf{B}_n(x) := q_n a_n + (1 - q_n) b_n, \tag{7}$$

$$\mathsf{B}_i(x) := q_i\bigl(a_i \oplus \mathsf{B}_{i+1}(x)\bigr) + (1 - q_i)\bigl(b_i \oplus \mathsf{B}_{i+1}(x)\bigr), \quad i = n-1, \dots, 2, \tag{8}$$

$$\mathsf{B}_1(x) := a_1 \oplus \mathsf{B}_2(x). \tag{9}$$

For $\varepsilon = (\varepsilon_2, \dots, \varepsilon_n) \in \{0, 1\}^{n-1}$ set

$$c_i(0) := a_i, \quad c_i(1) := b_i, \qquad w_i(0) := q_i, \quad w_i(1) := 1 - q_i.$$

Define

$$\mathsf{O}(x) := \sum_{\varepsilon \in \{0,1\}^{n-1}} \left(\prod_{i=2}^{n} w_i(\varepsilon_i)\right) \bigl(a_1 \oplus c_2(\varepsilon_2) \oplus \cdots \oplus c_n(\varepsilon_n)\bigr). \tag{10}$$

Theorem 3.1 (Finite-dimensional reduction). *For every $p > 1$, $n \ge 2$, and $0 < \ell \le u < \infty$,*

$$\Gamma_{p,n}(\ell, u) = \max_{x \in \mathcal{X}_n(\ell, u)} \frac{\mathsf{B}_1(x)}{\mathsf{O}(x)}. \tag{11}$$

In particular, the infinite-dimensional supremum over arbitrary product laws is attained by a product law for which D_1 is a point mass and D_i has support of size at most two for each $i \ge 2$.

Proof. The inequality "$\geq$" in (11) is immediate, since every law represented by (6) is admissible in (4). We prove the reverse inequality.

The product space $\mathcal{P}([\ell, u])^n$, with the product weak topology, is compact. The maps $D \mapsto B_i(D)$ are continuous by backward induction from (2), because all relevant integrands are continuous and bounded. The denominator in (4) is also continuous and is bounded away from zero, since

$$C_1 \oplus \cdots \oplus C_n \geq \left(n\ell^{-\alpha}\right)^{-1/\alpha} = \frac{\ell}{n^{p-1}}.$$

Thus the supremum in (4) is attained. Fix a maximizing product law D.

First fix $D_2, \ldots, D_n$. Let $\widehat{B}_2$ be the tail coefficient computed from those laws, and define

$$F(c) := c \oplus \widehat{B}_2, \qquad G(c) := \mathbb{E}_{D_2,\ldots,D_n}\left[c \oplus C_2 \oplus \cdots \oplus C_n\right].$$

Both functions are positive and continuous on $[\ell, u]$. For any first-coordinate law ν,

$$\frac{\int F(c)\, d\nu(c)}{\int G(c)\, d\nu(c)} \leq \max_{c \in [\ell,u]} \frac{F(c)}{G(c)}.$$

Replacing D_1 by a point mass at a maximizer of F/G therefore does not decrease the ratio. Since D was globally maximizing, the replacement is also globally maximizing. Hence there is a global maximizer with D_1 a point mass.

Now fix $k \in \{2, \ldots, n\}$ and suppose all coordinates except D_k have been fixed in a global maximizer. Define

$$f_k(c) := \begin{cases} c, & k = n, \\ c \oplus B_{k+1}, & k < n, \end{cases} \tag{12}$$

where B_{k+1} is the already fixed tail online coefficient. The numerator $B_1(D)$ depends on D_k only through the single scalar

$$m := \int f_k(c)\, dD_k(c).$$

Indeed $m = B_k$, and the earlier coefficients $B_{k-1}, \ldots, B_1$ are then determined by m and the fixed earlier laws.

The denominator is affine in D_k. More explicitly, there is a positive continuous function

$$g_k(c) := \mathbb{E}_{\{D_j : j \neq k\}}\left[C_1 \oplus \cdots \oplus C_{k-1} \oplus c \oplus C_{k+1} \oplus \cdots \oplus C_n\right]$$

such that the denominator is $\int g_k(c)\, dD_k(c)$.

Let $m_0 = \int f_k\, dD_k$. Consider the compact convex set

$$\mathcal{M}_k(m_0) := \left\{\nu \in \mathcal{P}([\ell, u]) : \int f_k(c)\, d\nu(c) = m_0\right\}.$$

On $\mathcal{M}_k(m_0)$ the numerator is fixed. Hence maximizing the ratio over $\mathcal{M}_k(m_0)$ is equivalent to minimizing the linear functional $\nu \mapsto \int g_k\, d\nu$. A minimizer exists and may be chosen at an extreme point of $\mathcal{M}_k(m_0)$.

We recall the elementary support bound for such extreme points. If an extreme point of $\mathcal{M}_k(m_0)$ had support containing three points at which f_k takes three distinct values, then one could choose three small disjoint neighborhoods of those points and assign nonzero signed masses to the restrictions of the measure on those neighborhoods so that both the total signed mass and the signed f_k-moment vanish. A sufficiently small positive and negative perturbation by this signed measure would remain in $\mathcal{M}_k(m_0)$, contradicting extremality. Since f_k is strictly increasing on $[\ell, u]$, an extreme point therefore has support of size at most two.

Thus D_k can be replaced by a law supported on at most two points while keeping B_k, and hence the numerator, unchanged and while not increasing the denominator. Since the starting law was globally maximizing, the denominator cannot strictly decrease under this replacement; otherwise the ratio would exceed the global maximum. Therefore the replacement is again globally maximizing.

Performing this replacement successively for $k = 2, \dots, n$ gives a global maximizer of the form (6). For such a law, (7)–(9) are exactly the online dynamic-programming coefficients and (10) is exactly the expected prophet value. This proves (11). The displayed maximum is attained because $\mathcal{X}_n(\ell, u)$ is compact and B_1/O is continuous. $\square$

Remark 3.2 (Extent of the reduction)**.** *Theorem 3.1 is exact, but for $n \geq 3$ it still leaves a nonconvex maximization over $3n-2$ real variables and a denominator containing 2^{n-1} terms. Thus it is partial progress on the general open problem rather than a closed-form solution for arbitrary n.*

4 The two-seller case

The finite-dimensional formula can be solved explicitly when $n = 2$.

Theorem 4.1 (Closed form for two sellers)**.** *Let $p > 1$, $0 < \ell \leq u < \infty$, and set*

$$\rho := \frac{u}{\ell}, \qquad \alpha := \frac{1}{p-1}.$$

Then

$$\Gamma_{p,2}(\ell, u) = \frac{1+\sqrt{\rho}}{2\sqrt{\rho}} \left(\frac{1+\rho^{\alpha/2}}{2} \right)^{1/\alpha}. \tag{13}$$

Equivalently,

$$\Gamma_{p,2}(\ell, u) = \frac{1+\sqrt{u/\ell}}{2\sqrt{u/\ell}} \left(\frac{1+(u/\ell)^{1/(2(p-1))}}{2} \right)^{p-1}. \tag{14}$$

If $\ell < u$, equality is attained by

$$C_1 = \sqrt{\ell u} \quad a.s.,$$

and

$$C_2 = \begin{cases} \ell, & \text{with probability } \dfrac{u-\sqrt{\ell u}}{u-\ell}, \\ u, & \text{with probability } \dfrac{\sqrt{\ell u}-\ell}{u-\ell}. \end{cases}$$

If $\ell = u$, the ratio is 1.

Proof. By homogeneity it suffices to take $\ell = 1$ and $u = r > 1$. Write

$$H(c,t) := c \oplus t = (c^{-\alpha} + t^{-\alpha})^{-1/\alpha}.$$

The finite-dimensional reduction gives $C_1 = c$ a.s. Since $B_2 = \mathbb{E}C_2$, the numerator is $H(c, \mu)$ where $\mu := \mathbb{E}C_2$.

For fixed c, the map $t \mapsto H(c,t)$ is strictly concave on $(0, \infty)$, because

$$\frac{\partial^2}{\partial t^2} H(c,t) = -(\alpha+1)c^{\alpha+1}t^{\alpha-1}(c^\alpha + t^\alpha)^{-1/\alpha-2} < 0. \tag{15}$$

Thus, among all laws on $[1, r]$ with mean μ, the expectation $\mathbb{E}H(c, C_2)$ is minimized by the endpoint law

$$\Pr[C_2 = 1] = \frac{r-\mu}{r-1}, \qquad \Pr[C_2 = r] = \frac{\mu-1}{r-1}.$$

Therefore

$$\Gamma_{p,2}(1,r) = \max_{c,\mu\in[1,r]} \frac{H(c,\mu)}{\frac{r-\mu}{r-1}H(c,1) + \frac{\mu-1}{r-1}H(c,r)}. \tag{16}$$

The following lemma evaluates this maximum. $\square$

Lemma 4.2 (Two-point Jensen-ratio calculation)**.** *For $r > 1$, $\alpha > 0$, and $H(c,t) = (c^{-\alpha} + t^{-\alpha})^{-1/\alpha}$,*

$$\max_{c,\mu\in[1,r]} \frac{H(c,\mu)}{\frac{r-\mu}{r-1}H(c,1) + \frac{\mu-1}{r-1}H(c,r)} = \frac{1+\sqrt{r}}{2\sqrt{r}}\left(\frac{1+r^{\alpha/2}}{2}\right)^{1/\alpha}. \tag{17}$$

The unique maximizer for $r > 1$ is $c = \mu = \sqrt{r}$.

Proof. For fixed $c \in [1,r]$, put

$$A := H(c,1), \qquad B := H(c,r), \qquad \Delta := B - A > 0,$$

and

$$R_c(\mu) := \frac{(r-1)H(c,\mu)}{(r-\mu)A + (\mu-1)B}.$$

The boundary values are $R_c(1) = R_c(r) = 1$, and by strict concavity of $H(c,\cdot)$, $R_c(\mu) > 1$ for $1 < \mu < r$.

The logarithmic derivative in μ is

$$\frac{\partial}{\partial\mu}\log R_c(\mu) = \frac{c^\alpha}{\mu(c^\alpha + \mu^\alpha)} - \frac{\Delta}{(r-\mu)A + (\mu-1)B}. \tag{18}$$

Multiplying by the positive quantity

$$\mu(c^\alpha + \mu^\alpha)((r-\mu)A + (\mu-1)B)$$

shows that the sign of (18) is the sign of

$$c^\alpha(rA - B) - \mu^{\alpha+1}\Delta.$$

Thus R_c has exactly one critical point in $(1,r)$, namely

$$\mu_c^{\alpha+1} = c^\alpha \frac{rA - B}{B - A}, \tag{19}$$

and this point is the unique maximizer of R_c over $[1,r]$.

It remains to maximize $M(c) := R_c(\mu_c)$ over $c \in [1,r]$. Write

$$P := (c^\alpha + 1)^{-1/\alpha}, \qquad Q := r(c^\alpha + r^\alpha)^{-1/\alpha}, \qquad d := \frac{Q}{P}.$$

Then $A = cP$, $B = cQ$, and $1 < d < r$. Also

$$d^\alpha = r^\alpha \frac{c^\alpha + 1}{c^\alpha + r^\alpha}, \tag{20}$$

so d is strictly increasing in c and

$$d < \sqrt{r} \quad\Longleftrightarrow\quad c < \sqrt{r}. \tag{21}$$

In the notation P, Q, d, the critical equation (19) becomes

$$\mu_c^{\alpha+1} = c^\alpha \frac{r-d}{d-1}. \tag{22}$$

By the envelope theorem, differentiating only the explicit c-dependence of $R_c(\mu)$ at $\mu = \mu_c$ gives

$$c\frac{d}{dc}\log M(c) = \frac{c^\alpha}{\mu_c(c^\alpha + \mu_c^\alpha)(Q-P)}\,\Psi(c), \tag{23}$$

where

$$\Psi(c) := rP - Q - (r-\mu_c)P^{\alpha+1} - (\mu_c - 1)Q^{\alpha+1}. \tag{24}$$

All factors outside $\Psi(c)$ in (23) are positive. We now determine the sign of $\Psi(c)$.

Using $Q = dP$ and $P^\alpha = (c^\alpha+1)^{-1}$, a direct rearrangement gives

$$\frac{c^\alpha+1}{P}\,\Psi(c) = T(\mu_c, d), \tag{25}$$

where

$$T(\mu, d) := (d-1)\mu^{\alpha+1} - (d^{\alpha+1}-1)\mu + d(d^\alpha - 1). \tag{26}$$

For fixed $d > 1$, the function $\mu \mapsto T(\mu, d)$ is strictly convex, and

$$T(1,d) = T(d,d) = 0.$$

Hence, for $\mu > 1$,

$$\operatorname{sgn} T(\mu, d) = \operatorname{sgn}(\mu - d). \tag{27}$$

It remains to compare μ_c and d.

Let $z := r/d$. Solving (20) for c^α and substituting into (22) yields

$$\left(\frac{\mu_c}{d}\right)^{\alpha+1} = \frac{\phi(d)}{\phi(z)}, \qquad \phi(t) := \frac{1-t^{-\alpha}}{t-1}, \quad t > 1. \tag{28}$$

The function ϕ is strictly decreasing on $(1,\infty)$: indeed,

$$\phi'(t) = \frac{\alpha t^{-\alpha-1}(t-1) - (1-t^{-\alpha})}{(t-1)^2},$$

and the numerator is negative because, after multiplying by $t^{\alpha+1}$, it becomes

$$(\alpha+1)t - \alpha - t^{\alpha+1} < 0, \qquad t > 1,$$

by strict convexity of $t \mapsto t^{\alpha+1}$. Therefore (28) implies

$$\mu_c > d \quad\iff\quad d < z = \frac{r}{d} \quad\iff\quad d < \sqrt{r}. \tag{29}$$

Combining (23), (25), (27), (29), and (21), we obtain

$$\frac{d}{dc}\log M(c) > 0 \quad \text{for } 1 < c < \sqrt{r}, \qquad \frac{d}{dc}\log M(c) < 0 \quad \text{for } \sqrt{r} < c < r.$$

Thus the unique maximizer is $c = \sqrt{r}$. Equation (22) then gives $\mu_c = \sqrt{r}$.

Finally let $s := \sqrt{r}$. Substitution into (16) gives

$$R_s(s) = \frac{H(s,s)}{\dfrac{s^2-s}{s^2-1}H(s,1) + \dfrac{s-1}{s^2-1}H(s,s^2)} = \frac{1+s}{2s}\left(\frac{1+s^\alpha}{2}\right)^{1/\alpha},$$

which is (17). $\square$

Completion of the proof of Theorem 4.1. The proof preceding Lemma 4.2 reduced the two-seller value with $\ell = 1, u = r$ to (16). Lemma 4.2 evaluates that maximum at $c = \mu = \sqrt{r}$. The worst second-coordinate law is the endpoint law with mean $\sqrt{r}$, namely

$$\Pr[C_2 = 1] = \frac{r - \sqrt{r}}{r - 1}, \qquad \Pr[C_2 = r] = \frac{\sqrt{r} - 1}{r - 1}.$$

Rescaling by ℓ gives the stated distribution and formulas (13)–(14). If $\ell = u$, every cost is deterministic and the ratio is 1. □

Corollary 4.3 (The quadratic two-seller formula)**.** *For $p = 2$,*

$$\Gamma_{2,2}(\ell, u) = \frac{1}{4}\left(\sqrt{\frac{u}{\ell}} + \sqrt{\frac{\ell}{u}}\right) + \frac{1}{2}.$$

Proof. When $p = 2$, one has $\alpha = 1$. Substituting $\alpha = 1$ into (13) gives

$$\frac{1 + \sqrt{\rho}}{2\sqrt{\rho}} \cdot \frac{1 + \sqrt{\rho}}{2} = \frac{(1 + \sqrt{\rho})^2}{4\sqrt{\rho}} = \frac{1}{4}\left(\sqrt{\rho} + \frac{1}{\sqrt{\rho}}\right) + \frac{1}{2}.$$

□

5 What is and is not resolved

Theorem 3.1 exactly removes the infinite-dimensional optimization over arbitrary product distributions. It shows that a worst case exists among distributions with the following simple form: the first seller has deterministic cost, and each later seller has a two-point cost distribution. This is a genuine reduction of the bounded no-in-house problem.

However, for $n \geq 3$ the theorem does not by itself give a closed-form competitive ratio. One must still solve

$$\max_{x \in \mathcal{X}_n(\ell, u)} \frac{\mathsf{B}_1(x)}{\mathsf{O}(x)},$$

which is generally nonconvex. Thus the contribution is best described as partial progress on the general open problem, together with a complete solution for $n = 2$ and arbitrary $p > 1$.

References

[1] Junjie Qin, Shai Vardi, and Adam Wierman. Minimization fractional prophet inequalities for sequential procurement. *Mathematics of Operations Research* 49(2):928–947, 2024. DOI: 10.1287/moor.2021.0173.

[2] Pranav Nuti. Open Problems in Operations Research, Problem 80. *Compute the exact competitive ratio for minimization fractional prophet inequalities without in-house production.* `https://pranav-nuti.github.io/open-problems-in-or/problem.html?id=W4379646686_p0&n=80&page=8`.